ALICE L WALTON SCHOOL OF MEDICINE

Respiratory: Coronaviridae (2026)

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Adapted from: Coronavirus Disease 2019

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Learning Objectives (5) Related (1) Versions

After completing this brick, you will be able to:

1. Describe the taxonomy of members of Coronaviridae.
2. Compare and contrast the diseases and epidemiology of the discussed coronaviruses.
3. Describe the life cycle of SARS-CoV-2.
4. Interpret lab 2.
5. Outline the treatment plan for infections caused by SARS-CoV-2.

CASE CONNECTION

A 67-year-old man presents to the emergency department with 5 days of fever, dry cough, and progressive shortness of breath. He reports recent exposure to his grandson who had a "cold." His past medical history is significant for hypertension and type 2 diabetes. On examination, his temperature is 38.4°C (101.1°F), blood pressure is 132/78 mmHg, heart rate is 104/min, respiratory rate is 26/min, and oxygen saturation is 88% on room air. Lung auscultation reveals bilateral fine inspiratory crackles (rales) at the lung bases without wheezing. Laboratory studies show lymphopenia and mildly elevated D-dimer. Chest imaging demonstrates bilateral, peripheral ground-glass opacities.

Which pathogen(s) do you suspect? What are the risk factors the patient possesses that increase the risk of complications? The patient was hospitalized. Which findings support this decision?

GO TO CONCLUSION ↓

Coronaviridae Taxonomy (LO1,2)

Coronaviruses are a large group of viruses that cause diseases ranging from the common cold to potentially more severe diseases like COVID-19. While recent headlines have been dominated by SARS-CoV-2, most coronaviruses are relatively harmless. However, outbreaks of novel coronaviruses have resulted in substantial morbidity ed with common cold coronaviruses.

SARS-CoV, MERS-CoV, and SARS-CoV-2 are all novel coronaviruses that have grab headlines in the last two decades and drastically increased mortality rates compared to common cold coronaviruses.

While the number of cases of SARS-CoV and MERS-CoV were relatively low, their mortality was disproportionally high. During the 2002-2003 outbreak of SARS-CoV, roughly 8096 cases were reported with 774 (9.6%) deaths. Since the first case of MERS, 2,494 cases have resulted in 858 (34.4%) deaths.

CLINICAL CORRELATION

SARS-CoV-2, the virus that causes COVID-19, has nowhere near the mortality rate of either SARS-CoV or MERS-CoV. However, what it lacks in mortality it makes up for in easy of transmission. While spread of SARS-CoV or MERS-CoV was limited to healthcare works, close family, and other close contacts, SARS-CoV-2 aggressively spreads via respiratory droplets.

As of July 2024, when global reporting slowed, over 775,000,000 cases of COVID-19 have been reported, with over 7 million deaths.

As with many epidemic or pandemic viral strains, SARS-CoV-2 is thought to be zoonotic, probably originating from bats or accidental release of the virus from a laboratory. The crossover to humans is believed to have occurred in late 2019 in a livestock market in Wuhan, China, but exactly how the initial human infection occurred is unclear.

Coronaviruses are enveloped and store their genetic information as (+)ssRNA. They also have a helical nucleocapsid. Coronaviruses receive their name due to the spiked protrusions on their capsid. It makes coronaviruses look like a crown; "corona" is latin for crown. The spiked protein that protrudes from the virus's surface bind host cells and facilitates entry (Figure 1).

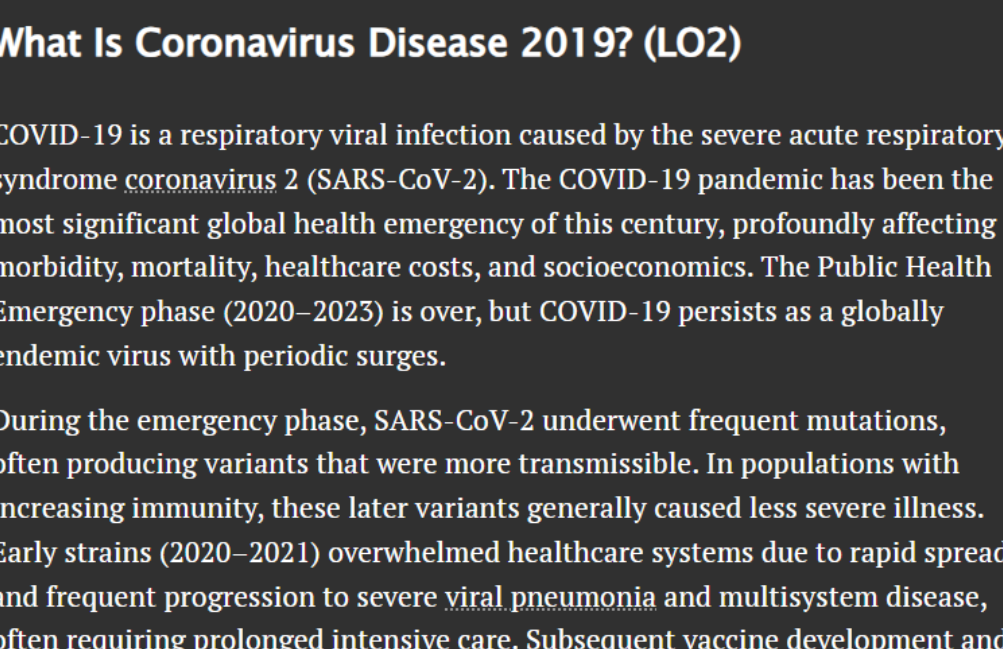

Figure 1: Coronavirus structure

What are some of the key characteristics of coronaviruses?

What Is Coronavirus Disease 2019? (LO2)

COVID-19 is a respiratory viral infection caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). The COVID-19 pandemic has been the most significant global health emergency of this century, profoundly affecting morbidity, mortality, healthcare costs, and socioeconomics. The Public Health Emergency phase (2020–2023) is over, but COVID-19 persists as a globally endemic virus with periodic surges.

During the emergency phase, SARS-CoV-2 underwent frequent mutations, often producing variants that were more transmissible. In populations with increasing immunity, these later variants generally caused less severe illness. Early strains (2020–2021) overwhelmed healthcare systems due to rapid spread and frequent progression to severe viral pneumonia and multisystem disease, often requiring prolonged intensive care. Subsequent vaccine development and targeted therapies, such as antivirals, have substantially improved outcomes by reducing critical illness and limiting transmission opportunities.

Risk Factors

The main risk factor tact with an infected person, particularly indoors or in poorly ventilated spaces. Transmission via contaminated surfaces is possible but considered rare.

Risk factors for severe COVID-19 illness include older age (≥50 years, but especially ≥65 years), hypertension, cancer, chronic lung disease, chronic kidney disease, diabetes (types 1 and 2), and immunocompromised states. Being unvaccinated or not up to date on vaccination also increases risk.

Figure 2 shows the COVID-19 death-risk ratio for select age groups and comorbid conditions. While none of these alone seems severe, when bundled together they may become significant in an individual patient.

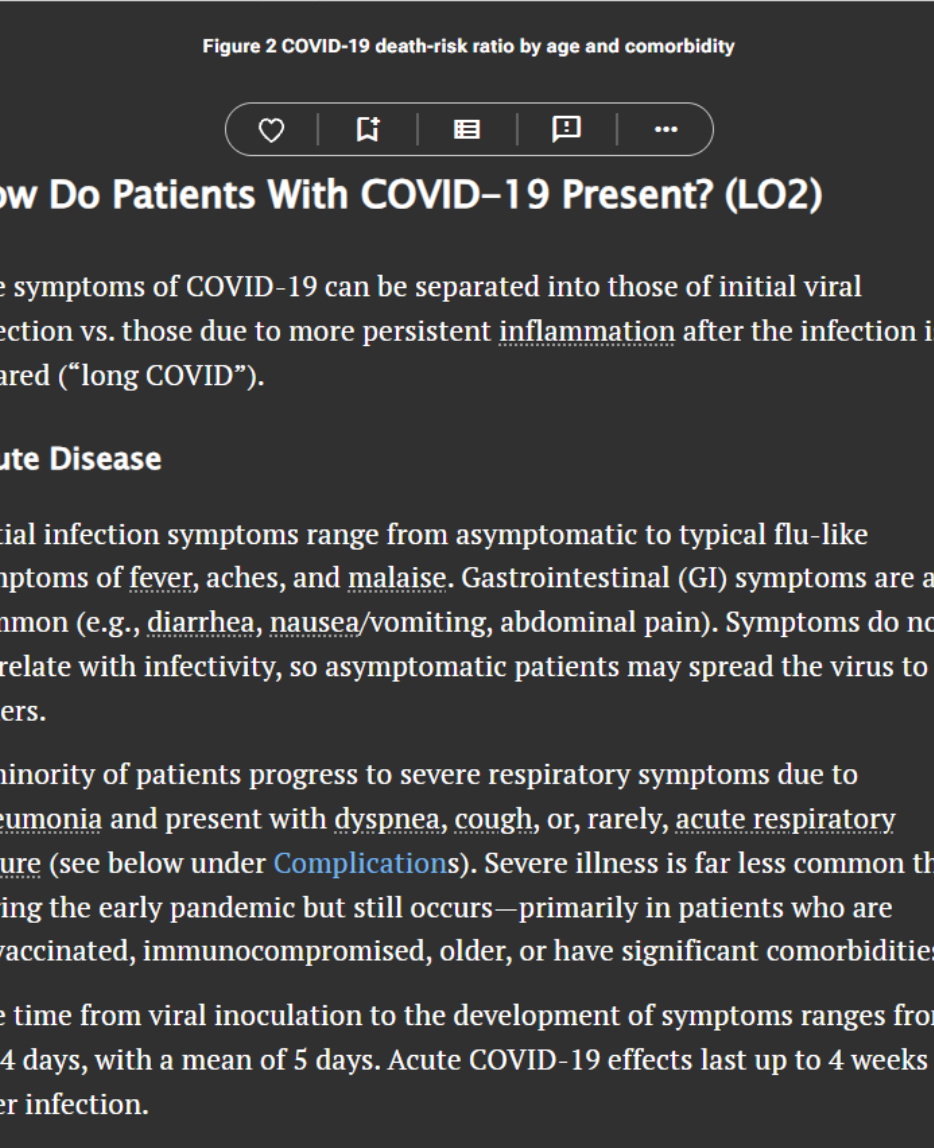

Figure 2 COVID-19 death-risk ratio by age and comorbidity

How Do Patients With COVID–19 Present? (LO2)

The symptoms of COVID-19 can be separated into those of initial viral infection vs. those due to more persistent inflammation after the infection is cleared ("long COVID").

Acute Disease

Initial infection symptoms range from asymptomatic to typical flu-like symptoms of fever, aches, and malaise. Gastrointestinal (GI) symptoms are also common (e.g., diarrhea, nausea/vomiting, abdominal pain). Symptoms do not correlate with infectivity, so asymptomatic patients may spread the virus to others.

A minority of patients progress to severe respiratory symptoms due to pneumonia and present with dyspnea, cough, or, rarely, acute respiratory failure (see below under Complications). Severe illness is far less common than during the early pandemic but still occurs—primarily in patients who are unvaccinated, immunocompromised, older, or have significant comorbidities.

The time from viral inoculation to the development of symptoms ranges from 2-14 days, with a mean of 5 days. Acute COVID-19 effects last up to 4 weeks after infection.

What is the incubation period for COVID-19?

Chronic Disease

Chronic symptoms may continue for more than 3 months after symptom onset. The mechanisms remain under investigation, but "long COVID" may affect 5%-6% of patients. Females are more affected than males (10% vs. 5%).

The most common symptoms of long COVID are fatigue, fever, dyspnea, and cough. Some patients develop a host of other symptoms and findings, likely due to **prolonged immune response** triggered by the viral infection. Such symptoms include nausea, vomiting, chest pain (from pericarditis), rashes, and leg swelling (from deep venous thrombi). Prolonged inflammation is suggested by persistent elevations in inflammatory markers like C-reactive protein and ferritin.

What symptoms prolonged immune responses?

Complications

COVID-19 can lead to complications affecting multiple organ systems. One contributing factor is the activation of neutrophil extracellular traps (NETs), which are part of the normal immune defense against pathogens. Neutrophils release webs of extracellular DNA and associated proteins that trap microbes and enhance their destruction. However, in COVID-19, these NETs become a double-edged sword, promoting a pro-inflammatory and a procoagulant state leading to multiorgan failure, in part due to activation of the coagulation cascade and consumption of clotting factors.

Complications include:

- **Pneumonia:** Some patients develop lower respiratory tract infection, progressing to viral pneumonia. This can impair gas exchange and cause dyspnea, hypoxemia, and, in severe cases, respiratory failure. In advanced disease, the lungs show diffuse inflammation with interstitial lymphocytic infiltrates and diffuse alveolar damage. Surfactant dysfunction may occur, contributing to alveolar collapse (atelectasis), as well as widespread inflammation and microvascular injury.
- **Respiratory fail** me (ARDS) occurs in patients with pne to the alveoli, reducing airflow and gas exchange. This often requires intubation and ventilation.
- **Cardiac effects:** these include heart failure, shock, and arrhythmias.
- **Neurologic effects:** these include motor and sensory disorders, seizures, and altered mental state (encephalopathy).
- **Thrombosis:** increased incidence of abnormal clotting can lead to deep vein thrombosis, pulmonary emboli, and strokes. The hypercoagulability is likely due to a combination of endothelial injury, NETs, and elevated fibrinogen causing increased viscosity.

Life Cycle and Pathogenesis (LO3)

Viral Transmission and Entry. COVID-19 and other coronaviruses spread mostly via **airborne particles** and **respiratory droplets**. Human-to-human transmission occurs with the release of respiratory secretions containing the virus (e.g., sneezing, laughing, and coughing). The virus can also be spread when people touch their eyes, mouth, or nose after contact with a contaminated surface.

In the upper respiratory system, the virus infects and destroys ciliated epithelial cells of the nasopharynx and upper airway, resulting in inflammation and impaired mucociliary clearance. When the virus reaches the distal respiratory tract, it infects type II pneumocytes via the receptor for **angiotensin-converting enzyme 2 (ACE2)**. There, it completes its virus life cycle. The virus can also infect and destroy capillary endothelial cells, leading to reduced blood perfusion of the alveoli.

Which receptor does SARS-CoV-2 use to enter cells and what type of respiratory cells do they infect?

Viral Propagation. SARS-CoV-2 translates many proteins that protect it from the initial innate immune response. For example, the virus inhibits the production of interferon-β by immune cells. Because COVID-19 is able to slow in innate immune response down, the high amount of virus produced increases transmission risk.

Acute Airway Inflammation. SARS-CoV-2 then becomes symptomatic by inducing airway inflammation, causing the familiar upper respiratory symptoms of rhinorrhea, nasal congestion, ear and sinus pain, sore throat, cough, and hoarseness. The inflammatory response is driven by cytokines. Macrophage activation and recruitment causes increased interleukin (IL)-1, IL-6, and tumor necrosis factor (TNF)-α. Please see the cytokine brick in the Immunology course for a refresher on cytokine functions.

Viral Mutation

RNA viruses have a higher rate of mutation compared to DNA viruses because of the reduced proofreading capacity of their polymerases. SARS-CoV-2 has evolved continuously, producing multiple variants with changes in transmissibility, immune escape, and treatment response. Over time, major variants have generally become more transmissible and, in populations with growing immunity, less likely to cause severe disease. This pattern favors viral persistence, as hosts remain infectious longer when illness is less severe.

For example, the Omicron variant had a higher rate of transmissibility than the Delta variant, probably because of a high number of mutations to its spike protein that increased its infective capability.

CLINICAL CORRELATION

COVID-19 has mutated rapidly since the original virus was identified. Most major variants are described by using the Greek alphabet. Alpha, Beta, Gamma, Delta, and Omicron are all variants of COVID-19 that have arisen. Each of these variants has mutations that alter the virus's virulence and helps it evade previous host and vaccine immune responses. Mutations in the spike glycoprotein are extremely common and are a cause of repeated infections with COVID-19.

SARS-CoV-2 variants circulating today are descendants of the Omicron lineage, with regional variation in dominance. The vaccine has been updated several times since its rollout to keep up with these variants and provide optimal protection from disease and hospitalization.

How Do We Dia

The history and physical exam in a symptomatic patient with COVID-19 is similar to that of other viral respiratory infections (upper and lower). A history of exposures to known COVID-19 is helpful.

On exam, patients may have fever. Those with pneumonia will have typical findings of viral pneumonia (e.g., fever, cough, and pulmonary crackles).

Tests for Viral Detection

Patients may arrive in the clinician's office already knowing they have COVID-19 due to widespread use of home antigen testing. The most commonly used tests to confirm COVID-19 infection are:

- **Antigen testing:** this immunoassay that detects antigens found on the surface of SARS-CoV-2 is widely available commercially and uses a nasal swab sample. Results are available in 15 minutes. While this test is specific, the lack of amplification of viral material makes it poorly sensitive—low viral loads may not trigger a positive result. Its primary value is in identifying high-viral-load patients likely to be contagious, which can guide patient behavior (e.g., self-quarantine, avoiding travel).
- **Polymerase chain reaction (PCR)/nucleic acid amplification test (NAAT):** this assay from a nasal swab is more sensitive and is the test most used in clinical settings. Using PCR means that even a small amount of viral RNA can be amplified to detectable levels. These tests are sometimes required for specific travel, employment, or immigration purposes and can be ordered by patients through a pharmacy.

Less used are assays that detect circulating immunoglobulin (Ig)G against SARS-CoV-2, typically ine. The antibodies often take weeks to d etect prior infection.

What is the diagnostic test of choice for COVID-19?

Tests for Patients with Long Covid

For patients being evaluated for chronic COVID illness, labs include:

- **Inflammation markers:** C-reactive protein, high ferritin, low fibrinogen
- **Liver:** liver enzymes (e.g., aspartate aminotransferase [AST] and alanine aminotransferase [ALT]), lactate dehydrogenase
- **Renal:** serum creatinine
- **Thrombosis:** D-dimer

How do we treat COVID–19? (LO5)

Treatment of SARS- verity, risk for progression, and timing of symptom onset. Unlike bacterial infections, antiviral therapy targets viral replication rather than cell wall or protein synthesis pathways. Most antiviral agents are most effective when administered early in the course of illness, when viral replication is highest. In contrast, patients with advanced disease often require anti-inflammatory therapies to reduce immune-mediated lung injury rather than direct antiviral therapy.

For non-hospitalized patients at high risk for progression to severe disease, oral antiviral therapy is recommended early after symptom onset. Nirmatrelvir–ritonavir is commonly used as first-line therapy. Nirmatrelvir inhibits the SARS-CoV-2 3CL protease, preventing cleavage of viral polyproteins required for replication. Ritonavir is included as a pharmacokinetic enhancer that inhibits CYP3A4 metabolism, increasing nirmatrelvir concentrations. This regimen significantly reduces hospitalization and mortality when initiated within the first 5 days of symptoms. Drug–drug interactions must be carefully evaluated due to ritonavir-mediated CYP inhibition.

Remdesivir, an intravenous nucleotide analog, is an alternative antiviral option. It inhibits RNA-dependent RNA polymerase, leading to premature termination of viral RNA synthesis. Remdesivir is commonly used for hospitalized patients requiring supplemental oxygen or as an alternative outpatient therapy when oral antivirals cannot be used.

For patients with progressive disease requiring supplemental oxygen, anti-inflammatory therapy becomes an essential component of treatment. Dexamethasone reduces inflammation by suppressing cytokine production and immune activation. Corticosteroids improve survival in patients requiring oxygen support but are not recommended for patients with mild disease who do not require oxyge mpair viral clearance.

CLINICAL CORRELATION

The timing of corticosteroid therapy in SARS-CoV-2 infection is critical because the dominant mechanism of disease changes over time. Early infection is driven primarily by **viral replication**, whereas later disease is driven largely by **host inflammatory injury**.

During the **early phase of infection**, viral replication is the primary driver of symptoms. The immune system plays an essential role in controlling viral spread through activation of innate immune responses, including interferon signaling and recruitment of antiviral immune cells. Administering corticosteroids during this phase can **suppress the immune response**, potentially delaying viral clearance and allowing continued viral replication. For this reason, corticosteroids are **not recommended in patients with mild disease who do not require supplemental oxygen**.

In contrast, patients with **severe disease requiring supplemental oxygen** often enter a later phase characterized by excessive immune activation, sometimes referred to as a **hyperinflammatory state** or cytokine-mediated lung injury. In this stage, inflammation rather than viral replication becomes the primary cause of tissue damage. Corticosteroids such as **dexamethasone** reduce the production of pro-inflammatory cytokines and limit immune-mediated lung injury. This reduces progression to respiratory failure and has been shown to improve survival in patients requiring oxygen or mechanical ventilation.

In patients with seve onal immunomodulatory therapy may be considered. Agents such as tocilizumab, an interleukin-6 (IL-6) receptor antagonist, reduce inflammatory signaling associated with cytokine-mediated lung injury. These therapies are typically used in combination with corticosteroids in patients demonstrating systemic inflammation.

Supportive care remains a cornerstone of therapy across all disease severities. This includes supplemental oxygen, fluid management, thromboprophylaxis when indicated, and close monitoring for complications such as respiratory failure or thrombosis. Overall, treatment of SARS-CoV-2 infection requires a stage-based approach, combining early antiviral therapy to suppress viral replication with later immunomodulatory therapy to reduce inflammatory injury in severe disease.

COVID–19 Prevention (LO2)

The main strategies for prevention of COVID-19 include various precautions and vaccination.

Precautions. Public health measures to reduce viral transmission remain important steps in the prevention of COVID-19. These include use of masks, handwashing, physical distancing (keeping at least 6 ft [2 m] away from others), avoiding nonessential gatherings, isolation, and quarantine.

These measures have been enforced by world governments and public health officials in varying ways, depending on local socioeconomics and the risk to the community. As the severity of COVID-19 has declined, fewer mandatory restrictions are in place in most countries, but this remains highly variable.

Vaccination. Vaccina incidence of symptomatic or severe COVID-19 disease. COVID-19 vaccines available in the United States are made of mRNA or protein. These vary in frequency of booster injections and duration of effect, but are generally recommended for everyone aged ≥6 years:

- mRNA vaccines deliver viral mRNAs to host cells, prompting host cells to produce the antigen and eliciting an immune response without harming the cells. Updated formulations are recommended to maintain protection against evolving variants.
- The protein subunit vaccine contains recombinant spike protein and adjuvant.

Worldwide, vaccine supplies are limited, and different countries (and populations) vary in their adherence to national vaccine guidelines. Overall, about 70% of the world population has been initially vaccinated, but this drops to 32% in low-income countries and remains under 25% in most African countries.

The percentage of people who have received boosters is much more variable and is less than 5% in many low-income countries. Variations also occur within higher-income countries like the US; these are largely cultural and often driven by misinformation about perceived risks and benefits of vaccination. Clinician education is therefore critical in increasing vaccination rates.

CASE CONNECTION BACK TO INTRODUCTION ↑

The dry cough and exposure to someone with cold-like symptoms suggests viral etiology. Common causes of viral pneumonia like influenza, RSV, and SAR-CoV-2 should be on you D sidered.

Diagnosis is confirmed with a nucleic acid amplification test (NAAT), such as RT-PCR, from a nasopharyngeal swab. The test is positive for SARS-Cov-2. Given his hypoxemia (SpO₂ 88% on room air) and risk factors for severe disease (age, diabetes), he requires hospital admission. Initial management includes supplemental oxygen to maintain saturation ≥92% and supportive care.

Summary

- Coronavirus disease 2019 (COVID-19) is a global pandemic that has placed a significant burden on the health care system and the economy.
- COVID-19 is caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), which crossed over from animals to humans.
- Risk factors for severe COVID-19 include advancing age and chronic diseases.
- Infection severity ranges from causing no symptoms (asymptomatic presentation) to causing acute respiratory failure and a catastrophic systemic inflammatory response.
- Initially, symptomatic patients present with upper respiratory symptoms that substantially overlap with the common cold.
- Some patients may develop pneumonia, dyspnea, and hypoxemia.
- COVID-19 enters cells through its spike glycoprotein binding the ACE-2 receptor. The ACE-2 receptor is found in the cells of the lungs, like type II pneumocytes.
- In addition to the respiratory tract, ACE-2 is expressed in the cardiovascular system, brain, kidneys, and gastrointestinal tract. This is why patients ma yond the respiratory system.
- SARS-CoV-2 is an enveloped, positive-sense, single-stranded RNA virus with a helical capsid.
- Transmission occurs primarily via airborne particles and respiratory droplets.
- COVID-19 is diagnosed by nucleic acid amplification test (NAAT) using reverse transcription polymerase chain reaction (RT-PCR).
- Prevention is mainly achieved by minimizing exposure to the disease through public health measures such as physical distancing and mask wearing.
- Vaccines effectively reduce the risk of infection, severity of disease, and risk of hospitalization.

Review Questions

1. Which of the following most accurately describes SARS-CoV-2?

A. ○ Double-stranded RNA virus with helical capsid
B. ○ Double-stranded RNA virus with icosahedral capsid
C. ○ Non-enveloped virus
D. ● Single-stranded RNA virus with helical capsid
E. ○ Single-stranded RNA virus with icosahedral capsid

Hide Explanation

The correct answer is single-stranded RNA virus with helical capsid (D). COVID-19 is caused by the SARS-CoV-2. The coronavirus genome is made up of a single-stranded RNA with a helical capsid. The coronavirus genome is single stranded, not double stranded (A and B), and has a helical capsid, not an icosahedral capsid (E). Non-enveloped virus (C) is incorrect because coronaviruses are enveloped.

2. Which of the following is least likely in a patient with prolonged inflammatory disease due to COVID-19 (aka long COVID)?

A. ○ Deep venous thrombosis
B. ○ Enlarged liver
C. ○ Fever
D. ○ Pericarditis
E. ● Thrombocytosis

Hide Explanation

The correct answer is thrombocytosis (E). Chronic inflammatory disease secondary to COVID-19 has multiple consequences, including cytopenias, not thrombocytosis. Thrombosis with DVT (A), hepatomegaly (B), fever (C) and chest pain due to pericarditis or myoca

3. Which receptor does SARS-CoV-2 use to enter host cells?

A. ○ CD4
B. ○ CD21
C. ● ACE2
D. ○ Sialic Acid Receptors
E. ○ Heparan Sulfate Proteoglycans

Hide Explanation

C) is correct. ACE2 is bound by SARS-CoV-2 and allows entry into target cells. A) CD4 is used by HIV to enter cells along with the coreceptors CXCR4 CCR5. B) EBV uses CD21 (CR2) to enter cells, D) Influenza virus uses sialic acid receptors, and E) HSV use heparan sulfate proteoglycans.

4. Which of the following tests is best for diagnosing COVID-19?

A. ○ Blood sample with antibody test
B. ○ Chest x-ray
C. ○ Nasophary
D. ● Nasopharyngeal swab with nucleic acid amplification test

Hide Explanation

The correct answer is nasopharyngeal swab with nucleic acid amplification test (D). The patient demonstrates symptoms of flu-like illness. These symptoms during the COVID-19 pandemic point toward a diagnosis of COVID-19. The diagnostic test of choice for COVID-19 is NAAT by RT-PCR. A blood sample with an antibody test (A) is not considered diagnostic for COVID-19 because of its limitations (e.g., low detection early in infection). A chest x-ray (B) can be normal during early disease, so it is not diagnostic. Findings are not distinct for COVID-19 so chest X-ray cannot distinguish between pathogens. Nasopharyngeal swab with antibody test (C) is incorrect because an antibody test is done on serum, not a nasopharyngeal swab.

How would you rate this Brick?

References

- COVID-19 deaths |
- Coronavirus Disease

References

- COVID-19 deaths | WHO COVID-19 dashboard
- Coronavirus Disease 2019 (COVID-19) | COVID-19 | CDC

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the taxonomy of members of Coronaviridae.~~
02 ☑ ~~Compare and contrast the diseases and epidemiology of the discussed coronaviruses.~~
03 ☑ ~~Describe the life cycle of SARS-CoV-2.~~
04 ☑ ~~Interpret laboratory findings to identify SARS-CoV-2.~~
05 ☑ ~~Outline the treatment plan for infections caused by SARS-CoV-2.~~

Objective Confidence: 5/5